# Cong Hoan Ngo

**Software Engineer**

*Hanoi, Vietnam*

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## Summary

Software Engineer with 7+ years building backend services, REST APIs, and web applications for US and Singapore businesses. Builds Python and Java backends (Flask, Spring Boot), including a move from Flask servers to serverless AWS (Lambda, API Gateway, S3) that cut infrastructure cost by about 25%, and web front ends in Angular and React. Since 2023 also builds LLM-powered pipelines and tools, and data platforms with Apache Airflow and dbt on Snowflake and BigQuery, with infrastructure as code (OpenTofu) and workloads on Kubernetes.

## Skills

**Programming:** Python, SQL, Java, JavaScript/TypeScript

**Software Engineering:** REST APIs, FastAPI, Flask, Spring Boot, React, Angular

**Cloud & Infrastructure:** AWS, GCP, Azure, OpenTofu/Terraform, Docker, Kubernetes, Helm, CI/CD

**AI & LLM:** OpenAI & Anthropic APIs, prompt engineering, LLM evaluation, embeddings & pgvector

**Databases & Warehousing:** Snowflake, BigQuery, PostgreSQL, MySQL, SAP HANA, Redis

**Data Engineering:** ETL/ELT, Apache Airflow, Airbyte, dbt, data modeling, data quality, web scraping

**BI & Analytics:** Metabase, Tableau

**Languages:** English - C2 Proficient (CEFR)

## Work Experience

### Pod Co.

**Data Engineer (Data & Software Engineering)** · Apr 2025 – Present · Remote · US-based company

- Built LLM-powered pipelines for trend and brand discovery: 100+ potential brands per week, over 30% became partnerships.
- Build internal web apps and tools for the team with React, such as a brand detection tool.
- Prototyped an LLM text-to-SQL analytics assistant on Snowflake with query guardrails and LLM evaluation.
- Maintain infrastructure as code with OpenTofu (Terraform) and monitor data workloads on Kubernetes (Helm).
- Build and operate ETL/ELT pipelines with Apache Airflow and Airbyte into Snowflake for Wholesale, B2B, and Distribution.
- Develop SQL data models in dbt on Snowflake and Metabase dashboards used daily by business, finance, and sales teams.
- Improve reliability and consistency of analytics on product performance, sales trends, and distribution metrics.

### VMO Holdings Technology JSC

**Data Engineer** · Dec 2023 – Apr 2025 · Vietnam

- Migrated legacy cron pipelines to Apache Airflow, improving scheduling, monitoring, and maintainability.
- Built Python ETL workflows for multi-source healthcare and pharma data in a data lake and warehouse (Azure, BigQuery).
- Supported critical data source migrations while preserving data integrity and downstream compatibility.
- Improved pipeline reliability and reconciliation, enabling analytics delivery for 200+ recurring reports.
- Investigated data discrepancies with analytics teams and delivered Tableau dashboards for business users.

### Maxflow R&D Co. Ltd.

**Software Engineer / Full-stack Developer** · Jan 2021 – Dec 2023 · Vietnam

- Moved Python Flask services to serverless AWS (Lambda, API Gateway, S3), cutting infrastructure cost by about 25%.
- Developed core features and REST APIs for an industrial simulation and analytics platform with configurable workflows.
- Implemented simulation modules for distance, energy, wear, and lifecycle cost across 100+ modeled scenarios.
- Built data preparation and feature extraction pipelines on Snowflake for predictive maintenance experiments.

### FPT Information System Corp.

**Back-end Developer** · Dec 2019 – Dec 2020 · Vietnam

- Built Java (Spring Boot) backend services for a national public procurement platform in a 5-engineer team.
- Contributed to Angular-based front-end improvements, reducing average page load time by approximately 20%.

### House3D Co. Ltd.

**Web Developer** · Feb 2019 – Dec 2019 · Vietnam

- Developed and maintained customer-facing web features for an interior design technology platform.

### Framgia Inc. / Sun* Inc.

**Web Developer Intern** · Jun 2018 – Sep 2018 · Vietnam

- Built a Ruby on Rails e-commerce project in a 3-member team with PayPal, BaoKim, and NganLuong payments.

## Projects

### Discode: AI Coding Assistants in Discord

**Personal Project · Python, PostgreSQL, Redis, Docker**

- Built a Discord bot platform that runs AI coding CLIs (Claude, Codex, OpenCode) as stateful, thread-based chat sessions.
- Designed a multi-process bot/runner/dispatcher on PostgreSQL and Redis with an outbox and idempotent job claims.

### Chatty Buddy: Multi-Provider AI Chat Client

**Personal Project · React, TypeScript, Zustand, Vite**

- Built a PWA chat client for OpenAI-compatible, Anthropic, and Ollama APIs in React 19, TypeScript, and Zustand.
- Implemented streaming adapters, tool permissions, and versioned backups; CI runs type checks, Vitest, and audits.

## Education

### VNU – Hanoi University of Science

**Bachelor of Science in Computer and Information Science** · 2016 – 2021 · Vietnam

- Coursework focus: Software Engineering, Database Systems, Data Structures and Algorithms.

## Certifications

**IBM** Data Architecture · May 2025

**Google** Associate Cloud Engineer · Dec 2024

**Google** Advanced Data Analytics · Nov 2024

**IBM** Data Engineering · Jul 2024

**Microsoft** Azure Data Engineering Associate · Jun 2024

**AWS** Cloud Solutions Architect · Jun 2024

## Honors & Awards

**Employee of the Year** (VMO Year-End Award, VMO Holdings) · 2024